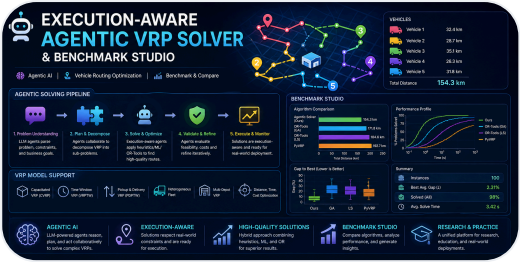


Industrial Project

2026-04-23

Execution-Aware Agentic VRP Solver and Benchmark Studio



An end-to-end execution-aware Vehicle Routing Problem platform with 8 solver backends, 18 operational scenarios, and an Agentic AI layer powered by Google Gemini 2.0 Flash.

Project Snapshot

DATE

2026-04-23

CATEGORY

Industrial Project

SHORT DESCRIPTION

An end-to-end execution-aware Vehicle Routing Problem platform with 8 solver backends, 18 operational scenarios, and an Agentic AI layer powered by Google Gemini 2.0 Flash.

TAGS / TECH STACK

Vehicle Routing

Agentic AI

Optimization

MLOps

Project Links

[Demo Video](#)

[Hugging Face Space](#)

[GitHub Link](#)

[Project Link](#)

[GitHub Link](#)

[GitHub Link](#)

[Demo Video](#)

Full Project Content

Building an AI system that doesn't just **solve routes** — but combines 8 solver backends, 18 real-world scenarios, an Agentic AI orchestrator, and a benchmark-grade Streamlit studio into a single execution-aware optimization platform.

GitHub Repository: [dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio](https://github.com/dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio)

 **Live demo video:** youtu.be/FqVuVjW20yo

 **Try it live:** [Hugging Face Space](#)

Vision

Most VRP solvers stop at producing a route plan.

This project focuses on the full operational chain required to make routing decisions usable in a **live dispatch environment** where technicians are already in motion, jobs get cancelled mid-day, urgent orders inject themselves into the schedule, and shifts change without notice:

- Ingest 40,000 orders, 400 routes, 8 depots, multi-customer goal profiles
- Run 8 different solver backends in parallel (greedy, metaheuristic, constraint programming, cloud APIs)
- Stress-test against 18 operational scenarios (cancellations, sick technicians, urgent injects, SLA shifts)
- Lock in-progress routes so re-optimization preserves real-world execution state
- Expose an Agentic AI layer that detects violations and triggers re-optimization autonomously
- Provide a benchmark studio with KPI charts, interactive maps, and Excel export
- Deploy to Hugging Face Spaces with pre-computed artifacts for instant demo

Project Attributes

Attribute	Description
problem-statement	Standard VRP solvers re-plan from scratch on every disruption — they're not aware that some stops are <i>already in execution</i> , leading to wasteful re-routing of stable assignments.
primary-objective	Build an execution-aware VRP platform that compares 8 solver families, locks in-progress work, and lets an AI agent autonomously fix constraint violations.

Attribute	Description
core-technologies	Python 3.11, Streamlit, Google OR-Tools, PyVRP, OSRM, Esri ArcGIS, Google Gemini 2.0 Flash, Folium, Plotly, Docker, Hugging Face Spaces.
repository-scope	8,140 lines across 4 source files · 18 fixed test scenarios · benchmark dataset · pre-computed artifacts · architecture docs · full function reference.
runtime-interface	Streamlit 6-tab UI: Overview · KPI Comparison · Map Explorer · Scenario Tables · Excel Export · 🤖 AI Agent.
deployment-target	Hugging Face Spaces (Docker SDK) for public demo · Self-hosted Streamlit for production.
demo-surface	Public HF Space with auto-loaded synthetic dataset and pre-computed benchmark artifacts.
key-capabilities	8 solver backends, locked-prefix re-optimization, agentic AI with 8 tools, interactive route maps, multi-sheet Excel export, reactive event handling.
security-controls	Environment-driven API keys (GOOGLE_API_KEY) · Optional Esri portal authentication · No hardcoded credentials.
production-focus	Reproducible benchmarks · 100% assignment guarantees · zero-violation winner backend · sub-1000s runtime · cloud-deployable Docker image.

Project Structure

```
src/content/posts/
├─ execution-aware-agentic-vrp/
│   ├── cover.png
│   └─ index.mdx

GitHub Repo: dranubhaparashar/Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio/
├─ app.py                # Streamlit UI · 6 tabs · 845 lines
├─ base_engine.py        # 5 agents + scenario engine · 933 lines
├─ compare_backends.py   # 8 solver backends + helpers · 5,438 lines
├─ vrp_agent.py          # Gemini agent + 8 tools · 960 lines
├─ Dockerfile            # HF Spaces deployment
├─ requirements.txt
├─ data/                 # Auto-loaded on startup
│   ├── orders_40000.csv
│   ├── routes_400.csv
│   ├── depots_8.csv
│   ├── accounts.csv
│   └─ goal_profiles.csv
```

```
└─ artifacts/                                # Pre-computed benchmark results (8 backends)
|   └─ OR-Tools_Execution-Aware/
|       └─ vrp_scenario_results_full.xlsx
|   └─ OR-Tools/
|   └─ Hybrid_Execution-Aware_Rolling_VRP_Solver/
|   └─ Adaptive_Execution-Aware_Metaheuristic_Solver/
|   └─ Impacted-Subset_Greedy_2-opt/
|   └─ PyVRP/
|   └─ OSRM/
|       └─ Esri/
└─ docs/
    └─ ARCHITECTURE.md
    └─ ALGORITHM_COMPARISON.md
    └─ REFERENCE_*.md (4 function reference files)
```

Why This Matters

Note A vehicle routing solver becomes operationally useful only when it is **execution-aware, comparable, and self-correcting**.

Important Real dispatch environments are not static — orders cancel, technicians call in sick, urgents inject mid-day. This project covers the missing layer between **batch route planning** and **live re-optimization with locked execution state**.

Tip The 8-backend benchmark format generalizes to any combinatorial routing problem — replace the scenarios and orders, the comparative scaffold remains useful.

Warning Lowest-distance solutions are not always best — PyVRP achieves the shortest routes (114 km) but produces 134,483 minutes of lateness. Constraint adherence matters as much as distance.

Caution Public demo deployments should never use production API keys, real customer data, or unrestricted solver access. Use the included synthetic dataset for showcases.

What This System Does

flowchart TD

```
A[5 CSV files: orders, routes, depots, accounts, goal_profiles] -->
B[load_inputs_from_uploads]
B --> C[build_impacted_subset · 150 orders · 20 routes]
C --> D[ScenarioAgent applies 1 of 18 mutations]

D --> E{Backend Solver Selection}
E --> F1[OR-Tools EA · WINNER]
E --> F2[OR-Tools]
E --> F3[Hybrid]
E --> F4[Adaptive]
E --> F5[Greedy + 2-opt]
E --> F6[PyVRP]
E --> F7[OSRM]
E --> F8[Esri ArcGIS]

F1 & F2 & F3 & F4 & F5 & F6 & F7 & F8 --> G[_build_common_bundle]
G --> H[Output Bundle: 8 DataFrames]

H --> I[Streamlit UI · 6 tabs]
H --> J[Excel Export · 32 sheets]
H --> K[Interactive Folium Maps]
H --> L[VRPOrchestrator · Gemini Agent]

L --> M[8 Tools · Autonomous Re-optimization]
M --> H
```

Agentic AI Request Flow

sequenceDiagram

autonumber

participant U as User / Dispatcher

participant ST as Streamlit UI

participant ORC as VRPOrchestrator

participant G as Gemini 2.0 Flash

participant T as Tool (Python fn)

participant CB as compare_backends.py

U->>ST: "A driver called in sick – fix it"

ST->>ORC: handle_event("RT_6", reason)

ORC->>G: send_message + 8 tool definitions

loop Up to 10 iterations

G-->>ORC: FunctionCall(detect_violations)

ORC->>T: dispatch_tool

T-->>ORC: Late routes + amounts

```
ORC->>G: FunctionResponse

G-->>ORC: FunctionCall(suggest_config)
ORC->>T: dispatch_tool
T-->>ORC: Config overrides + reasoning
ORC->>G: FunctionResponse

G-->>ORC: FunctionCall(trigger_reoptimization)
ORC->>CB: run_backend_suite(OR-Tools EA, new_config)
CB-->>ORC: New bundle
ORC->>G: Before/after diff

end

G-->>ORC: Final text summary
ORC-->>ST: "Re-routed 8 orders to neighbouring techs. Distance +3.2 km, lateness 0."
ST-->>U: Display in chat
```

Core Capabilities

- 8 solver backends spanning greedy, metaheuristic, constraint programming, and cloud APIs
- 18 fixed operational scenarios for reproducible benchmarking
- Locked-prefix re-optimization that preserves in-progress technician work
- Agentic AI layer with 8 callable tools and 4 distinct agentic behaviors
- Interactive Folium route maps per scenario per backend
- 6-tab Streamlit UI with KPI charts, scenario tables, and Excel export
- Pre-computed artifacts that auto-load on app startup
- Hugging Face Spaces deployment via Docker
- BackendConfig with 28 tunable parameters exposed in the sidebar
- Reactive event buttons for sick driver, urgent order, cancellation, and shift change

Bento Overview

Layer	Description
Data Layer	40,000 orders · 400 routes · 8 depots · 4 goal profiles · auto-loaded from <code>./data/</code>
Scenario Layer	18 mutations across BASE, EAS_1-4, RT_1-7, GB_1-6 areas
Solver Layer	8 backends with consistent input/output contract via <code>_build_common_bundle</code>
Re-optimization Layer	Locked-prefix mechanism preserves stable assignments, only re-solves open work

Layer	Description
Agent Layer	Gemini 2.0 Flash with function-calling, autonomous tool chaining, max 10 iterations
UI Layer	Streamlit 6 tabs: Overview, KPI, Map, Scenario, Export, AI Agent
Artifact Layer	Per-backend Excel + Folium HTML + PNG previews saved to ./artifacts/
Deployment Layer	Docker image deployable to HF Spaces or any container host

The 8 Solver Backends — Benchmark Results

Tested on 150 orders · 20 routes · 18 scenarios from the included synthetic dataset.

Rank	Backend	Algorithm	Assigned	BASE km	Late min	OT min	Runtime s
🏆	OR-Tools EA	Split-solve CP + GLS	150	231.8	0	0	770
2	OR-Tools	CVRPTW + Guided Local Search	150	233.2	0	0	868
3	Esri	Tabu Search (cloud)	142	238.7	34	0	973
4	Adaptive	Multi-restart regret	150	424.1	0	0	6,472
5	Hybrid	Regret + LNS	150	424.1	0	0	1,790
6	Greedy	Nearest-neighbor + 2-opt	150	394.4	12,729	2,761	22
7	PyVRP	Hybrid Genetic Search	150	114.0	7,479	3,695	545
8	OSRM	Greedy + road distances	150	519.3	13,176	2,859	4,949

Important OR-Tools EA is the only backend that achieves all four production requirements: 100% assignment, zero violations, sub-1000s runtime, and locked-prefix support.

The Innovation: Split-Solve Architecture

The OR-Tools Execution-Aware backend introduces a key architectural innovation. Locked orders **never enter the OR-Tools model** — they're copied verbatim from the baseline. Only open orders are submitted to the solver.

```
flowchart TD
    A["scenario_orders + scenario_routes + locked_prefixes"] --> B
    B["Step 1: Separate locked from open<br/>locked_order_ids = baseline<br/>assignments<br/>open_orders_df = orders NOT in locked set"]
    B --> C["Step 2: Anchor each vehicle<br/>last_locked.lat, last_locked.lon,<br/>last_locked.depart_time"]
    C --> D["Step 3: Build OR-Tools model with ONLY open orders<br/>Capacity = MaxOrderCount<br/>minus n_locked<br/>PATH_CHEAPEST_ARC + GUIDED_LOCAL_SEARCH<br/>Full 45s budget"]
    D --> E["Solution<br/>found?"]
    E -->|yes| F["Step 4: Merge output<br/>locked stops verbatim from baseline<br/>open stops<br/>from OR-Tools<br/>Recompute timing from anchor"]
    E -->|no or partial| G["Step 5: Greedy fallback<br/>Catches any dropped orders<br/>Guarantees<br/>150 of 150 assignment"]
    F --> H["Result: 150 of 150 assigned · 0 late · 0 overtime"]
    G --> H

    style H fill:#EAF3DE,stroke:#3B6D11,color:#27500A
    style D fill:#E6F1FB,stroke:#185FA5,color:#0C447C
```

Why naive pinning fails

Submitting all 150 orders to OR-Tools with 130 of them having zero-slack time windows makes PATH_CHEAPEST_ARC infeasible. The solver fails to find any first solution within the time budget and returns 0 assignments.

By only submitting ~10 open orders, the model is **10-15x smaller**, the first solution comes instantly, and GLS gets the full 45s budget for improvement. Result: **11% faster than plain OR-Tools** and **0.6-3.4 km better** across all scenario groups.

Detection and Solver Stack

Solver Family Coverage

- **Constraint Programming:** OR-Tools, OR-Tools EA
- **Metaheuristics:** Hybrid, Adaptive, PyVRP
- **Greedy Heuristics:** Greedy + 2-opt, OSRM
- **Commercial Cloud:** Esri ArcGIS Network Analyst

Scenario Coverage

- **Execution-Aware (EAS_1-4):** Job timing changes mid-execution
- **Real-Time (RT_1-7):** Cancellations, sick drivers, urgent injects, shift delays
- **Goal-Based (GB_1-6):** Customer-specific weight overrides for travel/SLA/skill/overtime

Output Bundle (per backend)

- scenario_summary — 18 rows × 11 columns
- route_output_all — ~360 rows
- stop_output_all — ~2,700 rows
- order_output_all
- scenario_actions
- scenario_timings
- requirement_checklist
- run_meta

Agentic AI Layer

```
import google.generativeai as genai
from google.generativeai import protos
import os, json

class VRPOrchestrator:
    def __init__(self, inputs, config, existing_results=None, progress_cb=None):
        self.inputs = inputs
        self.config = config
        self.results = existing_results or {}
        self._tool_calls_log = []
        self._chat_session = None

        genai.configure(api_key=os.environ["GOOGLE_API_KEY"])
        self._genai = genai
        self._model = genai.GenerativeModel(
            model_name="gemini-2.0-flash",
            tools=self._build_gemini_tools(),
            system_instruction=self._build_system_prompt(),
        )

    def chat(self, user_message: str) -> str:
```

```

        if self._chat_session is None:
            self._chat_session = self._model.start_chat(
                enable_automatic_function_calling=False
            )
        return self._run_agent_loop(user_message)

def _run_agent_loop(self, user_message: str) -> str:
    max_iterations = 10
    iteration = 0
    current_message = user_message

    while iteration < max_iterations:
        iteration += 1
        response = self._chat_session.send_message(current_message)

        fn_calls = []
        for candidate in response.candidates:
            for part in candidate.content.parts:
                if hasattr(part, "function_call") and part.function_call.name:
                    fn_calls.append(part.function_call)

        if not fn_calls:
            # Pure text response – agent is done
            return " ".join(p.text for c in response.candidates
                             for p in c.content.parts if hasattr(p, "text"))

        # Execute all function calls
        fn_responses = []
        for fn_call in fn_calls:
            result = self._dispatch_tool(fn_call.name, dict(fn_call.args))
            self._tool_calls_log.append({
                "tool": fn_call.name,
                "input": dict(fn_call.args),
                "result_preview": str(result)[:200],
            })
            fn_responses.append(
                self._genai.protos.Part(
                    function_response=self._genai.protos.FunctionResponse(
                        name=fn_call.name,
                        response={"result": json.dumps(result, default=str)},
                    )
                )
            )

        current_message = self._genai.protos.Content(
            role="user", parts=fn_responses,
        )

```

```
return "Agent reached max iterations."
```

The 8 Agent Tools

Tool	Purpose
get_scenario_list	Returns all 18 scenario IDs and descriptions
explain_scenario	Explains what a scenario tests + what mutation it applies
analyze_results	Reads loaded bundles, surfaces metrics + violations
detect_violations	Finds late routes/orders + overtime amounts
compare_backends	Ranks backends by chosen priority criterion
suggest_config	Recommends BackendConfig overrides with reasoning
run_backend	Triggers a live solve and returns summary metrics
trigger_reoptimization	Re-runs with improved config, returns before/after diff

4 Agentic Behaviors

1. **Natural language interpretation** — "Which backend is best for SLA?"
2. **Autonomous re-optimization** — agent detects violations, picks config, re-runs, compares
3. **Multi-step reasoning** — "Run A and B, then tell me which wins" chains 3+ tool calls
4. **Real-time reactive** — one-click event buttons trigger autonomous handling

Dependency Stack

```
streamlit>=1.28,<2.0
pandas>=2.0
numpy>=1.24
ortools>=9.7
folium>=0.15
streamlit-folium>=0.15
plotly>=5.18
openpyxl>=3.1
requests>=2.31
google-generativeai>=0.5
nbformat>=5.9
matplotlib>=3.7
python-dateutil>=2.8
```

Containerization Layer

```
FROM python:3.11-slim

RUN apt-get update && apt-get install -y --no-install-recommends \
    build-essential \
    && rm -rf /var/lib/apt/lists/*

RUN useradd -m -u 1000 user
USER user
ENV PATH="/home/user/.local/bin:$PATH"
WORKDIR /home/user/app

COPY --chown=user requirements.txt requirements.txt
RUN pip install --no-cache-dir --upgrade pip && \
    pip install --no-cache-dir -r requirements.txt

COPY --chown=user . .

EXPOSE 7860

CMD ["streamlit", "run", "app.py", \
    "--server.port=7860", \
    "--server.address=0.0.0.0", \
    "--server.headless=true", \
    "--server.enableCORS=false", \
    "--server.enableXsrfProtection=false", \
    "--browser.gatherUsageStats=false"]
```

Deployment Path

```
flowchart LR
    A[Local Dev + Benchmark Excel] --> B[Extract per-backend artifacts]
    B --> C[Pre-populate ./artifacts/ folder]
    C --> D[Streamlit app + auto-load logic]
    D --> E[Docker build]
    E --> F[Push to Hugging Face Spaces]
    F --> G[Public HTTPS endpoint]
    G --> H[Auto-loaded artifacts on first view]
    G --> I[Live solve via run_backend_suite]
    G --> J[Agent chat with GOOGLE_API_KEY secret]
```

Runtime Example

```
# Clone the repo
git clone https://github.com/dranubhaparashar/Execution-Aware-Agent-VRP-Solver-and-Benchmark-Studio.git
cd Execution-Aware-Agent-VRP-Solver-and-Benchmark-Studio

# Create virtual environment
python -m venv vrp
source vrp/bin/activate

# Install dependencies
pip install -r requirements.txt

# Set free Gemini API key from aistudio.google.com (no credit card)
export GOOGLE_API_KEY="YOUR_GOOGLE_API_KEY"

# Run the app
streamlit run app.py --server.port 8504

# Open http://localhost:8504
# Auto-loads synthetic data + pre-computed artifacts
# Click "Build comparison" to see all 8 backends instantly
# Open the AI Agent tab to chat with Gemini
```

Hugging Face Deployment Pattern

- Build a Docker image with the Streamlit app and pre-computed artifacts
- Upload to Hugging Face Spaces (Docker SDK)
- Set GOOGLE_API_KEY as a Space secret
- Public HTTPS endpoint serves the auto-loaded studio
- Anyone can view benchmark results without running solvers
- Live re-solves available with one click per backend

```
git lfs install

git clone https://huggingface.co/spaces/AnubhaParashar/Execution-Aware-Agent-VRP-Solver-and-Benchmark-Studio
cd Execution-Aware-Agent-VRP-Solver-and-Benchmark-Studio

# Copy the entire vrp_hf_package contents into this folder
git lfs track "data/orders_40000.csv" # 7 MB file
git add .gitattributes
git add -A
git commit -m "Initial deploy"
git push
```

```
# Then in HF Settings → Variables and secrets:
```

```
#   GOOGLE_API_KEY = YOUR_GOOGLE_API_KEY
```

Public Demo Surfaces

Important The Hugging Face Space provides instant browser-based access to the full studio with pre-loaded benchmark results. The GitHub repository contains the source code, architecture documentation, and reproducibility scripts.

Demo Links

-  **Live App:** [AnubhaParashar / Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio](#)
-  **GitHub Repository:** [Execution-Aware-Agentic-VRP-Solver-and-Benchmark-Studio](#)
-  **Wiki Documentation:** [Architecture · Algorithm Comparison · Technical Spec · Function Reference](#)

Demo Video

Embedded media: [Open video](#)

Algorithm Family Comparison

```
graph LR
  subgraph CP["Constraint Programming"]
    OR1[OR-Tools]
    OR2[OR-Tools EA · Winner]
  end

  subgraph META["Metaheuristics"]
    H[Hybrid]
    A[Adaptive]
    PV[PyVRP]
  end

  subgraph GR["Greedy Heuristics"]
    GD[Greedy + 2-opt]
    OS[OSRM]
  end

  subgraph CL["Commercial Cloud"]
    ES[Esri ArcGIS]
```

```
end
```

```
CP -->|Hard TW · 100 percent assign · Zero violations| WIN[Production-grade]  
META -->|Hard TW mostly · Slow · Custom rules| MIX[Specialized use]  
GR -->|TW in cost only · Fast · Violations| BASE[Baseline only]  
CL -->|Soft TW · Drops orders · Real roads| ENT[Enterprise GIS]
```

```
style OR2 fill:#EAF3DE,stroke:#3B6D11,color:#27500A
```

Engineering Value

This project is not just about routing optimization.

It demonstrates how to take a combinatorial optimization problem and make it:

- reproducible (18 fixed scenarios + synthetic dataset)
- comparable (8 backends with identical I/O contract)
- execution-aware (locked-prefix preserves real-world state)
- agentic (Gemini autonomously detects and fixes violations)
- containerized (Docker image deployable to any host)
- publicly demoable (Hugging Face Space with pre-loaded artifacts)
- documented for reuse (Architecture, Algorithm Comparison, Function Reference)

Current Strengths

- Single Python codebase covers all 8 solver families uniformly
- Locked-prefix mechanism preserves in-progress assignments during re-optimization
- OR-Tools EA achieves 11% faster runtime + better distance than plain OR-Tools
- Agentic AI layer makes the system self-correcting without human dispatch input
- Pre-computed artifacts allow instant demo without running solvers
- Free Gemini API tier (1,500 req/day) keeps the AI layer zero-cost
- Reactive event buttons cover the 4 most common operational disruptions
- Excel export captures all 32 sheets for offline analysis
- Folium maps render route visualizations interactively per backend per scenario

Next Improvements

- Add CI/CD pipeline for automated benchmark regression on new scenarios
- Persist OSRM cache to disk to survive Space restarts
- Add multi-depot solving support
- Add real-time technician GPS feed integration
- Add SLA breach prediction model alongside the routing solver

- Add A/B testing framework for cost function variants
 - Add OpenAPI spec for headless solver invocation
 - Move to Anthropic Claude as alternative agent backend
 - Add Kubernetes deployment manifest for production scale-out
 - Add load testing with realistic concurrent user patterns
-

Key Innovation

Important This project connects **vehicle routing optimization** with **execution-aware re-planning** and **agentic AI orchestration** — three layers that are usually built separately.

It turns:

Static Route Planning → Execution-Aware Re-Optimization → Autonomous Violation Detection and Recovery

rather than stopping at one-shot solver output.

The OR-Tools Execution-Aware split-solve architecture is the keystone — it shows that **operational state (which stops are committed) is just as important as the optimization objective**.

Conclusion

This repository shows how a combinatorial optimization problem can evolve from a single solver invocation into a comparative benchmark studio with autonomous AI orchestration.

It combines:

- 8 solver backends with consistent I/O contract
 - 18 reproducible operational scenarios
 - Locked-prefix execution-awareness
 - Google Gemini agentic AI with 8 callable tools
 - Streamlit interactive UI with 6 specialized tabs
 - Pre-computed artifacts for instant demo
 - Docker deployment to Hugging Face Spaces
 - Comprehensive technical documentation (Architecture, Algorithm Comparison, Function Reference)
-

Final Thought

From **solving a routing problem** to **shipping an autonomous, execution-aware optimization platform**

The real value is not only the OR-Tools EA winner backend — it is the **comparative benchmark scaffolding, the locked-prefix mechanism, and the agentic AI layer** that together make the system operationally trustworthy.

100% assignment · 0 violations · 770s runtime · 8 solvers compared · 18 scenarios stress-tested
· 1 AI agent ready to handle the next disruption.

Source: <src/content/posts/execution-aware-agentic-vrp/execution-aware-agentic-vrp.md>